CHAPTER 1

Statistics for Spatial Data

Perhaps, it may turn out a Sang;
Perhaps, turn out a Sermon.
ROBERT BURNS

Statistics, the science of uncertainty, attempts to model order in disorder. It is not surprising that students (and their teachers) find the subject enigmatic. However, as life experiences and scientific experiences accumulate, Statistics is usually recognized as an extremely powerful research tool. Even when the disorder is discovered to have a perfectly rational explanation at one scale, there is very often a smaller scale where the data do not fit the theory *exactly*, and the need arises to investigate the new, residual uncertainty.

Scientists and engineers have attempted to measure the level of disorder through, among other things, a quantity called *entropy*; see Jaynes (1957) and his subsequent writings (Rosenkrantz, 1983). Suppose there are $i = 1, \ldots, k$ possible states of nature that occur at random according to a probability distribution $(p_1, p_2, \ldots, p_k)$, where $0 \le p_i \le 1$, $i = 1, \ldots, k$, and $p_1 + p_2 + \cdots + p_k = 1$. Shannon (1948) defined entropy as

$$E \equiv -\sum_{i=1}^{k} p_i \cdot \log p_i,$$

where $x \cdot \log x \equiv 0$, for $x = 0$. When the state of nature i has a numerical interpretation, the mean $\mu \equiv \sum_{i=1}^{k} i \cdot p_i$ could be thought of as the center of the distribution and the variance $\sigma^2 \equiv \sum_{i=1}^{k} (i - \mu)^2 p_i$ as the amount of variability in the distribution. It is σ^2 that statisticians mostly use to quantify disorder, but not surprisingly E and σ^2 are closely related (see, e.g., Mukherjee and Ratnaparkhi, 1986).

There is a very interesting entropy inequality that can be proved. Consider the initial probability distribution $\mathbf{p}' \equiv (p_1, \ldots, p_k)$ evolving into a subsequent probability distribution $\mathbf{p}'A$, where A is a $k \times k$ matrix of nonnegative

elements each of whose rows sums to 1. Write $A = (a_{ij})$; then a_{ij} can be interpreted as the *conditional* probability that the subsequent state of nature is j *given* that the initial state of nature was i, and A is called a transition matrix. When states of nature evolve through repeated application of a stochastic mechanism described by the matrix A, the result is called a Markov chain (see, e.g., Isaacson and Madsen, 1976; Section 6.3). The entropy inequality is

$$E(\boldsymbol{\lambda}^{(1)} : \boldsymbol{\mu}^{(1)}) \geq E(\boldsymbol{\lambda}^{(0)} : \boldsymbol{\mu}^{(0)}),$$

where $E(\boldsymbol{\lambda}^{(l)} : \boldsymbol{\mu}^{(l)}) \equiv -\sum_{j=1}^{k} \lambda_j^{(l)} \cdot \log(\lambda_j^{(l)}/\mu_j^{(l)})$, $l = 0, 1$. Initially, $\lambda_j^{(0)} = p_j$ and $\mu_j^{(0)} = 1$, $j = 1, \ldots, k$, so that $E(\boldsymbol{\lambda}^{(0)} : \boldsymbol{\mu}^{(0)})$ is simply $E = -\sum_{j=1}^{k} p_j \cdot \log p_j$. The subsequent measures are $\lambda_j^{(1)} = \sum_{i=1}^{k} a_{ij}\lambda_i^{(0)} = \sum_{i=1}^{k} a_{ij} p_i$ and $\mu_j^{(1)} = \sum_{i=1}^{k} a_{ij}\mu_i^{(0)} = \sum_{i=1}^{k} a_{ij}$, $j = 1, \ldots, k$, depending on the transition matrix A.

A physical interpretation of this inequality might be that the universe tends to seek levels of entropy that are higher in relation to previous levels. Technology attempts to slow the *increase* in entropy by constraining evolving systems, but there are convincing arguments given that it will never decrease total entropy (e.g., Brooks and Wiley, 1988, Chapter 2). In short, Statistics is important because disorder is here to stay.

Independent-and-Identically Distributed-Data Model

Beginning classes in Statistics (and many of the more advanced ones) always assume that observations on a phenomenon are taken under identical conditions and that each observation is taken independently of any other. The data then form a random sample [i.e., are independent and identically distributed (i.i.d.)]; standard statistical techniques can be applied to build a statistical model and to estimate the model's parameters (see, e.g., Hogg and Craig, 1978). For example, Heyl and Cook (1936) describe experiments performed between May 1934 and July 1935 to determine the acceleration of gravity in a laboratory of the National Bureau of Standards, Washington, D.C. The method used was that of the reversible pendulum; various configurations of pendulum tube diameter and type of knife edge were used. One particular configuration, taken late May/early June 1934, yielded (*after* appropriate adjustments for flexure and clock rate were made)

$$76, 82, 83, 54, 35, 46, 87, 68,$$

expressed as deviations from $980{,}000 \times 10^{-3}$ cm/sec^2, in units of 10^{-3} cm/sec^2 (Heyl and Cook, 1936, Table 12). That these data can be modeled as arising from a random sample is probably a fair assumption, but a later experiment performed under a different configuration yielded deviations (again after appropriate adjustments)

$$76, 76, 78, 79, 72, 68, 75, 78,$$

in units of 10^{-3} cm/sec^2 (Heyl and Cook, 1936, Table 8). Again these could be modeled as a random sample, but because each experiment attempted to measure the same physical constant, the data should obviously be *combined* in some way. Is it fair to model the preceding 16 numbers as observations from a random sample?

Inhomogeneous-Data Model

Lack of homogeneity in data is usually accounted for in statistical models by a nonconstant-mean assumption; often the mean is assumed to be a linear combination of several explanatory variables. However, even after these large-scale variations are accounted for, there are often reasons to suspect inhomogeneous small-scale variations.

Cressie (1982) assumes the data from the experiment (to determine the acceleration of gravity) just described to be independent realizations from statistical distributions whose means are constant but whose *variances differ* (sometimes called heteroskedasticity) markedly, depending on the configurations of pendulum tube diameter and type of knife edge. Standard one-sample theory no longer applies, but it is still possible to construct a confidence interval for the common mean, based on a weighted t-like statistic.

The preceding example shows a relaxation of the identical-distribution assumption. Relaxation of the independence assumption is a further obvious way to generalize statistical models, but are these more general models of any scientific value? In the pages to follow, I hope to convince the reader that the answer to this question is very definitely "Yes."

Dependent-Data Model

Independence is a very convenient assumption that makes much of mathematical–statistical theory tractable. However, models that involve statistical dependence are often more realistic; two classes of models that have commonly been used involve intraclass-correlation structures and serial-correlation structures. These offer little scope for spatial data, where dependence is present in all directions and becomes weaker as data locations become more dispersed.

We have not yet been able to escape the three-dimensional world in which we live, nor the unidirectional flow of time through which we live. The notion that data close together, in time or space, are likely to be correlated (i.e., cannot be modeled as statistically independent) is a natural one and has been used successfully by statisticians to model physical and social phenomena. Purely temporal models, or time series models as they have come to be known (e.g., Box and Jenkins, 1970), are usually based on identically distributed observations that are dependent and occur at equally spaced time points. The unidirectional flow of time underlies the construction of these models.

Spatial models are a more recent addition to the Statistics literature. Geology, soil science, image processing, epidemiology, crop science, ecology,

forestry, astronomy, atmospheric science, or simply any discipline that works with data collected from different spatial locations, need to develop (not necessarily statistical) models that indicate when there is dependence between measurements at different locations. However, the models need to be more flexible than their temporal counterparts, because past, present, and future have no analogy in space, and furthermore it is simply not reasonable to assume that spatial locations of data occur regularly, as do most time series models.

When dealing with data where (space–time) dependencies are likely, two approaches can be contrasted. Departures from the independence paradigm could be modeled, or statistical procedures could be constructed that would be robust to these departures. Throughout this book I shall consider mostly the modeling approach.

The Weather

It is useful to consider an example that has both the temporal and spatial component in it. The weather is a universal topic of conversation (some would say it is a resort of the desperate), and among its various facets that of rainfall is paramount. To the more than one million inhabitants of South Australia, the driest state in the driest continent, drought can be devastating. In a series of papers, Cornish, with others (1936, 1954, 1958, 1961, 1976), looked at temporal and spatial aspects of 26 rainfall recording stations in South Australia, shown in Figure 1.1. Data analyzed were monthly rainfall amounts at 26 locations that had been recording for a period of 30 years or more. Thus, at any one location, the data form a typically lengthy time series (e.g., for Adelaide, the capital city of South Australia, there were 1248 monthly observations available), and at any one time point, the data are spatial and not more than 26 in number; over time and space they form a collection of approximately 20,000 observations. More recently, meteorological space–time data sets have been collected for the purposes of studying the effects of atmospheric pollution, in particular *acid rain* (see e.g., Peters and Bonelli, 1982). Daily data collection over a number of years at various locations throughout say the northeast United States, yields a massive data set. But most of it is temporal so that, spatially speaking, the data are still rather sparse. Nevertheless, spatial prediction is just as important as temporal prediction, because people living in those cities and rural districts without monitoring stations have the same right to know how little or how much their water or their air is polluted. (Section 4.6 discusses some of the issues surrounding acid rain and analyzes a spatial data set.)

Scope and Purpose

The topics covered in this book are almost exclusively related to data analysis and statistical modeling of spatial data; the basic components are spatial locations $\{\mathbf{s}_1, \ldots, \mathbf{s}_n\}$ and data $\{Z(\mathbf{s}_1), \ldots, Z(\mathbf{s}_n)\}$ observed at those locations. Usually the data are assumed random and sometimes the locations are

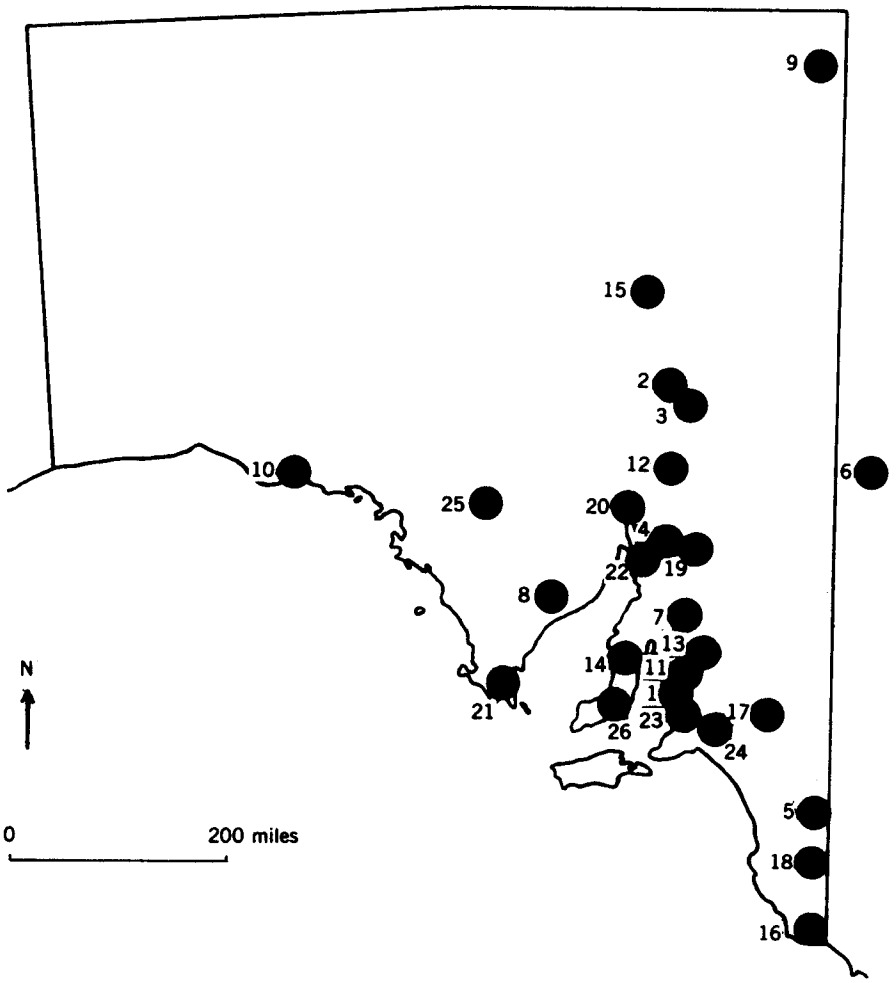


Figure 1.1 Map of South Australia showing 26 rainfall recording locations: Adelaide (1), Beltana (2), Blinman (3), Booleroo Centre (4), Bordertown (5), Broken Hill (6), Clare (7), Cleve (8), Cordillo Downs (9), Fowler's Bay (10), Gawler (11), Hawker (12), Kapunda (13), Maitland (14), Marree (15), Mount Gambier (16), Murray Bridge (17), Naracoorte (18), Peterborough (19), Port Augusta (20), Port Lincoln (21), Port Pirie (22), Stirling West (23), Strathalbyn (24), Yardea (25), Yorketown (26).

assumed random. Moreover, once the locations are given, the possibility of mistaken or imprecise positioning is generally not modeled. Some modifications to this paradigm are occasionally considered; for example, see Sections 4.4, 4.5, and 6.1.

Issues regarding the measurement, storage, and retrieval of spatial information are extremely important, although that is not the emphasis in this book. A geographic information system (GIS) is a collection of computer

software tools that facilitate, through georeferencing, the integration of spatial, nonspatial, qualitative, and quantitative data into a data base that can be managed under one system environment (e.g., Burrough, 1986). Much of the research in GIS has been concerned with computational geometry, spatial discretization, spatial languages and user interfaces, systems designs and architectures for data integration, spatial data handling approaches for alternative system architectures (such as parallel processing and neural networks), and so forth. Satellites are amassing huge amounts of data, a small fraction of which is being analyzed. This embarrassment of riches must be controlled by being selective; selectivity can be guided by the types of problems solved and the models that will be needed to solve them. Geographic information systems have only recently begun to incorporate model-based spatial analyses into their information-processing subsystems; it is hoped that this book will provide ideas for further initiatives.

Another topic that receives little attention in this book is the definition of "distance" between two sites at known locations. Mostly, I represent the two sites as vectors $\mathbf{s} = (s_1, \ldots, s_d)'$ and $\mathbf{u} = (u_1, \ldots, u_d)'$ in Euclidean space $\mathbb{R}^d$ and use the Euclidean distance, $\|\mathbf{s} - \mathbf{u}\| \equiv \{\sum_{i=1}^{d}(s_i - u_i)^2\}^{1/2}$, although in Section 4.6 great-arc distances are chosen instead. In any particular problem, other distances may be more appropriate, such as road miles or a measure that combines, say, travel expense, convenience, and time. These alternative metrics lead naturally to questions of *clustering* (e.g., Mardia et al., 1979, Chapter 13). For example, which regions are "close" on a "map" drawn according to the new metric? Based only on pairwise "distances" between sites, multidimensional scaling (e.g., Kruskal and Wish, 1977; Mardia et al., 1979, Chapter 14) constructs a geometric configuration of the site "locations." A danger in the uncritical use of clustering methods for spatial data analysis is that important spatial contiguities may be violated; Legendre (1987) advocates constraining the clustering to respect these contiguities.

My principal purpose for writing this book is to make statistical theory and methods for spatial data available to scientists and engineers. I believe that the full potential of Statistics is seen in its application to substantive problems, and I have tried to reinforce this in the applications I have given throughout the book. I hope that readers will find ideas and information here that will be useful in their own areas of scientific endeavor, although I do recognize that there are large parts that may be inaccessible to all but the most theoretically inclined. Such sections have been marked with an asterisk to warn the nonstatistician to obtain help in reading them. By way of balance, those sections that are truly oriented toward applications have been marked with a dagger.

The book is divided into three parts, dealing with spatial processes indexed over continuous space (a topic sometimes referred to as geostatistics), spatial processes indexed over lattices in space (the spatial analogue of time series), and spatial point processes (including marked point processes and

random set processes). These are convenient subareas that do not exhaust all of the statistical models one might use to analyze spatial data. The next section gives some background on spatial problems and defines the general model considered in this book.

1.1 SPATIAL DATA AND SPATIAL MODELS

The first manifestations of statistics for spatial data appear to have arisen in the form of data maps. For example, Halley (1686) superimposed, onto a map of land forms, directions of trade winds and monsoons between and near the tropics, and attempted to assign them a physical cause.

Spatial models appeared much later. For example, Student (1907) was concerned with the distribution of particles throughout a liquid. Instead of analyzing their spatial positions he aggregated the data into counts of particles per unit area: A hemocytometer of area 1 mm^2, divided into 400 squares, was used to count yeast cells. Student found that the distribution of the number of cells per square followed a Poisson distribution. (Daley and Vere-Jones, 1988, p. 8, can be consulted for a brief history of counting problems.)

R. A. Fisher was clearly aware of spatial dependence in agricultural field experiments, because he went to such great lengths to remove it (see Fisher, 1935, Chapter IV). In the 1920s and 1930s, at Rothamsted Experimental Station in England, he established the principles of randomization, blocking, and replication. As well as controlling for unwanted bias, randomization also neutralizes (but does not remove) the effect of spatial correlation (Yates, 1938; Section 5.6.2). However, it should be realized that randomization does *not* neutralize the spatial correlation at spatial scales larger or smaller than the plot dimensions.

Fairfield Smith (1938) was concerned with choosing plot dimensions so that any increase in plot size would yield little decrease in error variance. Although his analysis was empirical, the very formulation of the problem recognizes the presence of spatial correlation in field experiments. Models for such phenomena did not begin to appear until much later (Whittle, 1954).

Nearest-neighbor methods for analyzing agricultural field trials attempt to take spatial dependence into account, indirectly, by using residuals from neighboring plots as covariates, or by differencing (Papadakis, 1937; Bartlett, 1938, 1978; Wilkinson et al., 1983; Besag and Kempton, 1986). A review of these methods is given in Section 5.7.1.

Presently, Statistics can be found in any quantitative discipline, giving rise to rich variations on the theme that was played at Rothamsted 60 years ago. In areas such as geology, ecology, and environmental science, it is not often possible (nor always appropriate) to randomize, block, and replicate the data. There is a need for new statistical models and approaches that address new questions arising from old and new technologies. Many of the resulting

problems, such as resource assessment, environmental monitoring (e.g., for global warming), and medical imaging, are spatial in nature.

The General Spatial Model

Statistics, in all its guises from exploratory data analysis to asymptotic distribution theory of parameter estimators, relies on a more or less vague stochastic model. I shall present such a model for spatial data, which has a very simple structure that is flexible enough to handle an extremely large class of problems (including the ubiquitous i.i.d. case). The data may be continuous or discrete, they may be spatial aggregations or observations at points in space, their spatial locations may be regular or irregular, and those locations may be from a spatial continuum or a discrete set. At the very least, the stochastic model is used to summarize extant data or to predict unobserved data. It may or may not explain why a particular phenomenon occurs, and so should be distinguished from the more commonly accepted use of the word "model." To many scientists, a model must have causative dynamic components, but that is not necessarily the case in this book.

Let $\mathbf{s} \in \mathbb{R}^d$ be a generic data location in d-dimensional Euclidean space and suppose that the potential datum $\mathbf{Z}(\mathbf{s})$ at spatial location $\mathbf{s}$ is a random quantity. Now let $\mathbf{s}$ vary over index set $D \subset \mathbb{R}^d$ so as to generate the multivariate random field (or random process)

$$\{\mathbf{Z}(\mathbf{s}): \mathbf{s} \in D\}; \tag{1.1.1}$$

a realization of (1.1.1) is denoted $\{\mathbf{z}(\mathbf{s}): \mathbf{s} \in D\}$. This "superpopulation" model for spatial data is used exclusively throughout the book, although brief discussion of a design-based approach (whose randomness is derived from sampling the deterministic process $\{z(\mathbf{s}): \mathbf{s} \in D\}$) is given in Section 2.3.

Usually, D is assumed to be a *fixed* (i.e., nonrandom) subset of $\mathbb{R}^d$ (see, e.g., Vanmarcke, 1983), but I shall assume more generally that D is a *random set*. Formally speaking, a random set is a measurable mapping from a probability space onto a measure space of (usually closed) subsets of $\mathbb{R}^d$ (see Section 9.3 for more details). Less formally speaking, I shall assume that D as well as $\mathbf{Z}$ may vary from realization to realization, giving another source of randomness to the problem. This simple structure allows me to talk about problems with continuous spatial index, problems with lattice index, spatial point patterns, and more, all under the same umbrella. Thus, the three seemingly distinct parts of this book alluded to earlier treat special cases of (1.1.1), where the index set D could be a random set. Specifically:

Part I (Chapters 2 through 5): Geostatistical data. D is a fixed subset of $\mathbb{R}^d$ that contains a d-dimensional rectangle of positive volume; $\mathbf{Z}(\mathbf{s})$ is a random vector at location $\mathbf{s} \in D$.

Part II (Chapters 6 and 7): Lattice data. D is a fixed (regular or irregular) collection of countably many points of $\mathbb{R}^d$ (this can be generalized to

where D is a graph in $\mathbb{R}^d$; see Section 6.1); $\mathbf{Z}(\mathbf{s})$ is a random vector at location $\mathbf{s} \in D$.

Part III (Chapter 8): Point patterns. D is a point process in $\mathbb{R}^d$ or a subset of $\mathbb{R}^d$; $\mathbf{Z}(\mathbf{s})$ is a random vector at location $\mathbf{s} \in D$. In its most general form this generates a *marked spatial point process*; when no $\mathbf{Z}$ is specified the usual spatial point process is obtained [or one could think of scalar $Z(\mathbf{s}) \equiv 1$, for all $\mathbf{s} \in D$].

Part III (Chapter 9): Objects. D is a point process in $\mathbb{R}^d$; $Z(\mathbf{s})$ is *itself* a random set. This yields processes such as the Boolean model (Section 9.4) and a related cell-growth process (Section 9.7).

The flexibility of (1.1.1) is now apparent, and although it is clear that D and $\mathbf{Z}(\mathbf{s})$ could be even more general random quantities, the fact that D is a subset of $\mathbb{R}^d$ allows (1.1.1) to be called a *spatial process*. It is even possible to think of the ubiquitous i.i.d. (possibly multivariate) model as a 0-dimensional spatial process, because any spatial index is unimportant. I have not mentioned how the D process and the $\mathbf{Z}$ process might covary. For most of this book, one or the other is fixed so the question does not arise. When both are random, they are usually assumed to be independent, or one process is analyzed conditionally upon the observed values of the other. Spatial modeling occurs then within the $\mathbf{Z}$ process (Parts I and II), within the D process, or within both processes (Part III), and typically involves modeling the large- and the small-scale variations in terms of a finite number of parameters.

I prefer to keep (multivariate) *time-series* processes separate from spatial processes by using the index t and denoting them as

$$\{\mathbf{Z}(t): -\infty < t < \infty\}. \tag{1.1.2}$$

The unidirectional flow of time sometimes forces one to distinguish between (1.1.2) and (1.1.1) in $\mathbb{R}^1$ (see Section 6.3).

A *space–time process* will be denoted as

$$\{\mathbf{Z}(\mathbf{s}; t): \mathbf{s} \in D(t), t \in T\}, \tag{1.1.3}$$

where each of $\mathbf{Z}$, D, and T is possibly random. This is as general a process as will be considered in this book.

Description, Indication, and Estimation

As disciplines in Statistics mature, their statistical analyses advance through three stages of development that I describe as *description*, *indication*, and *estimation*. For the first, the goal is to summarize the data (perhaps to suggest models). For the second, estimates of model parameters are obtained from the data, but no measures of precision are available. For the third, enough distribution theory is at hand to allow (approximate) inference on the model parameters; for example, the bias and variance of an estimator can be

calculated and estimated. Statistics for spatial data is still a young discipline, showing increasing signs of maturity. I hope this book will contribute to, and be an influence on, its formative years.

1.2 INTRODUCTORY EXAMPLES

In this section, I shall present examples where data are spatial in nature. Some data sets are new and some have already appeared in the literature, although they may not have been analyzed spatially. It is not my intention to present spatial statistical analyses in this section, but rather to illustrate the range of problems that can be addressed. For some of these data, more complete analyses are carried out later in the book.

As described in Section 1.1, spatial data can be thought of as resulting from observations on the stochastic process

$$\{\mathbf{Z}(\mathbf{s}): \mathbf{s} \in D\}, \tag{1.2.1}$$

where D is possibly a random set in $\mathbb{R}^d$. I shall present some special cases that are typical of spatial statistical problems one might encounter.

1.2.1 Geostatistical Data

Geostatistics (Matheron, 1962, 1963a, 1963b) emerged in the early 1980s as a hybrid discipline of mining engineering, geology, mathematics, and statistics. Its strength over more classical approaches to ore-reserve estimation is that it recognizes spatial variability at both the large scale *and* the small scale, or in statistical parlance it models both spatial trend *and* spatial correlation. Trend-surface methods (e.g., Whitten, 1970) include only large-scale variation, assuming independent errors. Watson (1972) compares the two approaches and points out that most geological problems have a small-scale variation, typically exhibiting strong positive correlation between data at nearby spatial locations. One of the most important problems in geostatistics is to predict the ore grade in a mining block from observed samples. Matheron (1963b) has called this process of prediction *kriging* (see Chapter 3).

Exploration data on coal ash (Gomez and Hazen, 1970, Tables 19 and 20) for the Robena Mine Property in Greene County, Pennsylvania, are an example of regularly spaced data in $\mathbb{R}^2$, although some locations have not been sampled or their observations are missing. Section 2.2 performs a spatial exploratory data analysis on these mining data, and kriging is carried out in Section 3.4.

The geostatistical method has found favor among soil scientists who seek to map soil properties of a field from a small number of soil samples at known locations throughout the field. For example, soil pH in water, soil

electrical conductivity, exchangeable potassium in the soil, soil–water tension, and soil–water infiltration are some of the variables that could be sampled. Water erosion is of great concern to agriculturalists, because rich topsoil can be carried away in runoff water. Some forms of tillage result in greater soil–water infiltration than others, and the study described in Section 4.3 investigates this question. The greater the infiltration, the less the runoff, resulting in less soil erosion and less stream pollution by pesticides and fertilizers. Also, greater infiltration implies better soil structure that is more conducive to crop growth. In Cressie and Horton (1987) and Section 4.3, it is explained how double-ring infiltrometers were placed at regular locations in a field that had received four tillage treatments. From these data the spatial relationships can be characterized, treatments compared, and kriging maps drawn.

There are many published sources for case studies using geostatistics, although the original data are usually not given for confidentiality reasons (original data for both of the examples just introduced are available in Sections 2.2 and 4.3, respectively). The books by David (1977, 1988), Journel and Huijbregts (1978), and Clark (1979) are all devoted to applications in the mining industry. Application of geostatistics in other areas abound; for example, in rainfall precipitation (e.g., Ord and Rees, 1979), atmospheric science (e.g., Thiebaux and Pedder, 1987), soil mapping (e.g., Burgess and Webster, 1980), and predicting groundwater contaminant concentrations (e.g., Istok and Cooper, 1988). For further applications, see the various sections of Chapter 4 that contain analyses of data sets from hydrology, soil science, public health, uniformity trials, and acid rain.

Geostatistical-type problems are distinguished most clearly from lattice- and point–pattern-type problems by the ability of the spatial index $\mathbf{s}$ to vary *continuously* over a subset of $\mathbb{R}^d$. That is not to say that methods from one class of problems cannot be borrowed from methods usually associated with another class; see, for example, Sections 4.4 and 7.6, where the same public-health data set is analyzed using two approaches.

1.2.2 Lattice Data

A *lattice* of locations evokes an idea of regularly spaced points in $\mathbb{R}^d$, linked to nearest neighbors, second-nearest neighbors, and so on. In this book these will be referred to as *regular* lattices, allowing for the possibility of *irregular lattices*, whose relative displacements do not follow a predictable pattern and whose linkages are not always obvious from their geometry. Of all the possible spatial structures that (1.2.1) can generate, a data set whose spatial locations are a regular lattice in $\mathbb{R}^d$ is the closest analogue to a time series observed at equally spaced time points.

Remote sensing from satellites offers an efficient means of data gathering. For example, it allows information to be gathered rapidly on weather patterns, mineral distribution, and crop acreage without having to conduct

lengthy and labor-intensive traditional surveys. Satellites orbit the earth and receive data in the form of electromagnetic reflectance waves at a number of frequencies, including those in the visible part of the spectrum. By various sampling and integration methods (see Section 7.4 for more details), the earth's surface is divided into small rectangles (e.g., 56 m × 56 m) called *pixels* (short for picture elements). An agricultural scene of interest (around, say, 34,000 km^2) has certain proportions devoted to wheat, corn, soybeans, and so forth that need to be estimated. These various crops have their own reflectance properties that, together with noise, are remotely sensed. Thus the data are received as a regular lattice in $\mathbb{R}^2$ (neglecting the earth's curvature) and are identified with the centers of their respective pixels. (This is analogous to temporal problems where economic time-series data, such as yearly export earnings, are aggregated over the whole year but are identified with single, equally spaced points on the time axis.) There is a large overlap between remote-sensing techniques and (low-level) medical-imaging techniques; although the spatial scales are vastly different, the form of the data and the questions being asked are often similar. Statistical models for such data need to express the fact that observations nearby (in time or space) tend to be alike; see Section 7.4.

Sometimes it might be expected that nearby observations tend to be dissimilar. Competition between plants for light and soil nutrients could lead to large healthy plants being surrounded by less sturdy ones. Mead (1967) analyzes data on cabbages and investigates this competition effect.

In contrast to geostatistical problems, data from lattice problems may be exhaustive of the phenomenon. Of course, sampling may also occur; for example, suppose only a small window of the full data set is available.

There have been various published studies based on lattice data (although such studies have not always exploited the spatial component) that include Mercer and Hall (1911) (see also Sections 4.5 and 7.1), Batchelor and Reed (1918), Cochran (1936), Bartlett (1974), Cliff and Ord (1981), Ripley (1981), Symons et al. (1983) (see also Sections 4.4, 6.2, and 7.6), and Cressie and Chan (1989).

1.2.3 Point Patterns

Point patterns arise when the important variable to be analyzed is the location of "events." Most often the first question to be answered is whether the pattern is exhibiting complete spatial randomness, clustering, or regularity. For example, consider the locations of longleaf pines in a natural forest in southern Georgia (given in Section 8.2). What is the biological significance of clustering of these trees? The variable, diameter at breast height, is also recorded along with the tree's location. Do large (small) trees cluster and how do large and small trees interact? The size variable is usually called a *mark variable*, and the whole process is then called a *marked spatial point process*.

But the mark variable does not have to be a real variable; it could, for example, be a set. The process then consists of $\{Z(\mathbf{s}_i)\colon \mathbf{s}_i \in D\}$, D a spatial point process and $Z(\mathbf{s}_i)$ a *random set* (Section 9.3) located at $\mathbf{s}_i \in D$. Data from this process are often observed as a realization of

$$X = \cup\{Z(\mathbf{s}_i)\colon \mathbf{s}_i \in D\}, \tag{1.2.2}$$

which is referred to as a (generalized) Boolean model (Serra, 1980) or a mosaic process (Diggle, 1981b). The data are often images, and their assimilation to a model like (1.2.2) requires higher levels of image analysis than the methods considered in Section 7.4 of Part II. The goal is to estimate parameters of the random set *and* the point process; see Cressie and Laslett (1987) and Section 9.5 where an artificially generated Boolean model of random parallelograms is analyzed. Diggle (1981b) analyzes the incidence of heather according to a Boolean model; see also Section 9.5. A model for tumor growth that iterates the Boolean model through successive stages is developed in Section 9.7. Pictures of cell islands growing *in vitro* are presented and analyzed in a manner that takes *shape* as well as size into account.

Published studies involving analyses of spatial point patterns can be found *inter alia* in Pielou (1959, 1977), Matern (1960), Getis and Boots (1978), Marquiss et al. (1978), Ripley (1981), Diggle (1983), and Upton and Fingleton (1985). Of course, there is a large overlap of methods for patterns occurring in space and for those occurring in time; the reader interested in *temporal* methods is referred to Cox and Lewis (1966), the articles in Lewis (1972) (in particular the review by Daley and Vere-Jones, 1972), Cox and Isham (1980), and Daley and Vere-Jones (1988). A second-moment measure of spatial point patterns is adapted for the temporal animal-behavior data in Section 8.4.

1.3 STATISTICS FOR SPATIAL DATA: WHY?

Some simple spatial models will be given to show the effect of correlation on estimation, prediction, and design. The models allow closed-form expressions to be calculated, from which discussion of more general issues can be initiated.

Estimation

Consider the following simple statistical model, taught in all beginning Statistics service courses. Suppose $Z(1), \ldots, Z(n)$ are independent and identically distributed (i.i.d.) from a Gaussian (i.e., normal) distribution with unknown mean μ and known variance σ_0^2. The minimum-variance unbiased

estimator of μ is

$$\bar{Z} \equiv \sum_{i=1}^{n} Z(i)/n, \tag{1.3.1}$$

and inference on μ is straightforward: The estimator $\bar{Z}$ is Gaussian with mean μ and variance σ_0^2/n. Thus, a two-sided 95% confidence interval for μ is

$$\left(\bar{Z} - (1.96)\sigma_0/\sqrt{n}\,, \bar{Z} + (1.96)\sigma_0/\sqrt{n}\right). \tag{1.3.2}$$

Instead of independent data, now suppose the data are positively correlated with a correlation that decreases as the separation between data increases:

$$\operatorname{cov}(Z(i), Z(j)) = \sigma_0^2 \cdot \rho^{|i-j|}, \qquad i, j = 1, \ldots, n, 0 < \rho < 1. \tag{1.3.3}$$

It is well known that such a correlation function results from a first-order autoregressive process $Z(i) = \rho Z(i-1) + \epsilon(i)$, $i \geq 1$, where $\epsilon(i)$ is part of an i.i.d. sequence of Gaussian random variables with zero mean and variance $\sigma_0^2(1-\rho^2)$ and is independent of $Z(i-1)$; see, e.g., Fuller (1976, Section 2.3). However, that is *not* the generating mechanism I have in mind; here, $\mathbf{Z} \equiv (Z(1), \ldots, Z(n))'$ are *spatial* data in $\mathbb{R}^1$ from which prediction of $Z(0)$ or $Z(3/2)$ is just as appropriate as that of $Z(n+1)$.

Now,

$$\begin{aligned}\operatorname{var}(\bar{Z}) &= n^{-2}\left\{\sum_{i=1}^{n}\sum_{j=1}^{n} \operatorname{cov}(Z(i), Z(j))\right\} \\ &= \{\sigma_0^2/n\}\big[1 + 2\{\rho/(1-\rho)\}\{1 - (1/n)\} \\ &\qquad -2\{\rho/(1-\rho)\}^2(1-\rho^{n-1})/n\big].\end{aligned} \tag{1.3.4}$$

For $n = 10$ and $\rho = 0.26$, $\operatorname{var}(\bar{Z}) = \{\sigma_0^2/10\}[1.608]$, and a two-sided 95% confidence interval for μ is

$$\left(\bar{Z} - (2.485)\sigma_0/\sqrt{10}\,, \bar{Z} + (2.485)\sigma_0/\sqrt{10}\right).$$

Thus, failure to realize the presence of positive correlation in the data leads to a confidence interval (1.3.2) that is *too narrow*; for $n = 10$ and $\rho = 0.26$, the actual coverage probability of (1.3.2) is 87.8%, not 95%. [If $-1 < \rho < 0$, (1.3.2) would be too wide, but because positive spatial dependence is seen more often, these introductory remarks concentrate on situations where ρ is positive.]

Some intuitive understanding of the effect of spatial correlation can be obtained from (1.3.4). Write it as

$$\operatorname{var}(\bar{Z}) = \sigma_0^2/n', \tag{1.3.5}$$

where

$$n' \equiv n\Big/\big[1 + 2\{\rho/(1-\rho)\}\{1 - (1/n)\} - 2\{\rho/(1-\rho)\}^2(1-\rho^{n-1})/n\big] \tag{1.3.6}$$

can be interpreted as the *equivalent number of independent observations* (see Section 4.6.2). If $n = 10$ and $\rho = 0.26$, then $n' = 6.2$; that is, 6 independent observations achieve approximately the same precision as 10 correlated observations [correlated according to (1.3.3) with $\rho = 0.26$]. For large n, $n' \simeq n/[(1+\rho)/(1-\rho)]$, showing that correlation has an effect on inferences even in large samples.

Spatial models, more complicated than (1.3.3), show the same general behavior. Haining (1988) considers constant-mean Gaussian models in $\mathbb{R}^2$ that are conditionally specified autoregressions (Section 6.6), and simultaneously specified autoregressions and moving averages (Section 6.7.1), each with an unknown variance parameter σ^2. He compares the variance of $\bar{Z}$ assuming independence, with the variance of $\bar{Z}$ assuming positive dependence; he also compares the latter with the variance of the maximum likelihood estimator $\hat{\mu}$ of the constant mean μ. In general, classical inference based on $\bar{Z}$ and $\hat{\sigma}^2/n$ is misleading; for positive spatial dependence, $\widehat{\text{var}}(\bar{Z})$ and $\widehat{\text{var}}(\hat{\mu})$ are typically much larger than $\hat{\sigma}^2/n$.

Grenander (1954) demonstrates that, for a class of time-series models that includes (1.3.3), $\bar{Z}$ and the maximum likelihood estimator

$$\hat{\mu} \equiv \left\{Z(1) + (1-\rho)\sum_{i=2}^{n-1} Z(i) + Z(n)\right\}\Big/\{n - (n-2)\rho\}$$

(which is also the minimum variance unbiased estimator) have the same asymptotic (100%) efficiency as $n \to \infty$. Does this contradict the preceding calculations? No, because there, inference for μ was based on $\bar{Z}$ *and* σ_0^2/n, and was found to be invalid. Grenander's result allows one to conclude that inference based on $\bar{Z}$ and σ_0^2/n' [n' is given by Eq. (1.3.6)] is not only valid but is also asymptotically *efficient*. General models for which this happens are given by Grenander and Rosenblatt (1984, Section 7.4), and they include stationary autoregressive time series.

Prediction

Suppose now that an unknown observation $Z(n+1)$ is to be *predicted* from data $\mathbf{Z} \equiv (Z(1), \ldots, Z(n))'$, where it is assumed $Z(1), \ldots, Z(n), Z(n+1)$ are jointly Gaussian, identically distributed with unknown mean μ and known variance σ_0^2, and *independent*. From Section 3.2, the predictor $p(\mathbf{Z}; n+1)$ that satisfies the unbiasedness condition $E(p(\mathbf{Z}; n+1)) = \mu$ and minimizes

the mean-squared prediction error

$$E(Z(n+1) - p(\mathbf{Z}; n+1))^2 \tag{1.3.7}$$

is

$$p_0(\mathbf{Z}; n+1) = \bar{Z}, \tag{1.3.8}$$

the sample mean. The minimized mean-squared prediction error is

$$ms_0 = \sigma_0^2\{1 + (1/n)\}. \tag{1.3.9}$$

When the independence assumption is replaced with (1.3.3), the unbiased predictor that minimizes the mean-squared prediction error (1.3.7) is

$$p_\rho(\mathbf{Z}; n+1) = \rho Z(n) + (1-\rho)\frac{\{Z(1) + (1-\rho)\sum_{i=2}^{n-1} Z(i) + Z(n)\}}{\{n - (n-2)\rho\}}; \tag{1.3.10}$$

see Section 3.2. Abbreviate (1.3.10) to $p_\rho(\mathbf{Z})$. Notice that $\rho = 0$ in (1.3.10) yields (1.3.8). If $\bar{Z}$ is used instead of (1.3.10), the mean-squared prediction error is

$$E(Z(n+1) - \bar{Z})^2 = \sigma_0^2\Big\{1 + (1/n)\Big[1 + 2\{\rho/(1-\rho)\}\{\rho^n - (1/n)\} - 2\{\rho/(1-\rho)\}^2(1-\rho^{n-1})/n\Big]\Big\}. \tag{1.3.11}$$

For $n = 10$ and $\rho = 0.26$, (1.3.11) is $\sigma_0^2\{1.09051\}$. Compare this to (1.3.9), which is $\sigma_0^2\{1.1\}$: The difference is not large. Indeed, when spatial correlations decay geometrically with distance, classical prediction intervals are often approximately *valid*. However, they can be highly *inefficient*.

Assuming the dependence model given by (1.3.3), the minimized mean-squared prediction error is obtained by calculating (1.3.7) with optimal predictor (1.3.10). That is,

$$ms_\rho = \sigma_0^2\Big[1 - \rho^2 + \{(1-\rho)^2(1+\rho)/(n - (n-2)\rho)\}\Big]; \tag{1.3.12}$$

see Section 3.2. Notice that $\rho = 0$ in (1.3.12) yields (1.3.9).

For $n = 10$ and $\rho = 0.26$, (1.3.12) is $\sigma_0^2\{1.01952\}$. Thus a 95% prediction interval for $Z(n+1)$, based on the optimal predictor (1.3.10), is

$$\left(p_\rho(\mathbf{Z}) - (1.96)\sigma_0\{1.01952\}^{1/2}, p_\rho(\mathbf{Z}) + (1.96)\sigma_0\{1.01952\}^{1/2}\right). \tag{1.3.13}$$

This should be compared to the *wider* 95% prediction interval for $Z(n+1)$ based on $\bar{Z}$:

$$\left(\bar{Z} - (1.96)\sigma_0\{1.09051\}^{1/2}, \bar{Z} + (1.96)\sigma_0\{1.09051\}^{1/2}\right). \quad (1.3.14)$$

For large n, the squared ratio of the prediction-interval width from using $p_\rho(\mathbf{Z})$ to that from using $\bar{Z}$ is approximately $1 - \rho^2$, which is a measure of the (asymptotic) efficiency of inference based on $\bar{Z}$ versus inference based on the optimal predictor $p_\rho(\mathbf{Z})$. Thus, for $\rho = 0.26$ the efficiency measure is 93% and for $\rho = 0.5$ it is 75%.

Experimental Design

Another example of the importance of exploiting spatial correlation is in the area of experimental design. Suppose that experimental units are laid out in a $t \times b$ array made up of b blocks (columns), each with t units in them. Units that are most alike are put in the same block so that all of the gross heterogeneity occurs between blocks. Suppose further that t treatments are to be compared as follows. A response variable is measured after applying the treatments to experimental units in such a way that each treatment appears exactly once in each block. A design (i.e., an allocation of treatments to units) of the type just described is known as a *complete block design*.

In papermaking, toward the end of the process when the paper is still a continuous sheet in a roll (that is typically 10 ft wide and 3×10^4 ft long), the sheet is *calendered*. A calender is a stack of two or more hard smooth rollers through which the paper is passed, whose purpose is to reduce the caliper of the sheet (so making it more dense) and increase the smoothness of the sheet. In calendering, high compressive stresses are used to make a more compact, thinner sheet with a smoother surface. In terms of basic smoothness and thickness, most of the variation is across the width of the paper roll. Therefore, any experiment on calender treatments (e.g., number and degree of smoothness of calender rolls, temperature of the sheet) has to take this into account, as well as the possible spatial correlation running the length of the sheet.

A well designed calendering experiment in the laboratory should split the sheet into several narrower sheets (e.g., each of width 2 ft), allowing these to form the blocks of the experiments. Within the blocks, pages closer together will tend to be more alike (in terms of gloss or caliper) than those further apart, and this can be modeled with (one-dimensional) spatial correlation. Therefore, when a calender treatment is applied to a "page," its location within the block should be useful information for performing an efficient analysis.

Let the response on the ith unit of the jth block be $Y_{ij(k)}$, $i = 1, \ldots, t$, $j = 1, \ldots, b$, $k = 1, \ldots, t$, where k denotes the treatment assigned (according to the design) to that unit. Assume an additive model

$$Y_{ij(k)} = \mu + \alpha_k + \beta_j + \delta(i, j), \quad (1.3.15)$$

where $\delta(\cdot, \cdot)$ is a Gaussian error process with zero mean. Further, assume independence of the δs between blocks but spatial dependence (1.3.3) down blocks. That is,

$$\operatorname{cov}(\delta(i,j), \delta(i',j')) = \begin{cases} 0, & j \neq j', \\ \sigma^2 \rho^{|i-i'|}, & j = j', \end{cases} \tag{1.3.16}$$

where $|\rho| < 1$.

A goal of experimental design is to find the allocation of treatments to units that will give the most precise estimates of (estimable) treatment effects, $\tau_k \equiv \alpha_k - (\sum_{l=1}^{t} \alpha_l / t)$, $k = 1, \ldots, t$. Specifically, one might minimize

$$A \equiv \sum\sum_{1 \leq k < l \leq t} \operatorname{var}(\hat{\tau}_k - \hat{\tau}_l) \Big/ \binom{t}{2}, \tag{1.3.17}$$

where $\{\hat{\tau}_k\colon k = 1, \ldots, t\}$ are the generalized-least-squares estimators of the treatment effects (see Section 5.6.2). Gill and Shukla (1985a) find that *first-order nearest-neighbor (NN) balanced* designs, whose end units satisfy further balance conditions, are optimal designs. First-order NN balanced means that the number of times two treatments appear adjacent in a block is constant for every possible pair of treatments. For example, consider $t = 10$ and $b = 5$; then Table 1.1 shows an assignment of treatments $0, 1, \ldots, 9$ within the five blocks that is a first-order NN balanced design.

Let A_O, A_N, and A_R denote the value of (1.3.17) for, respectively, the optimal design, the design in Table 1.1, and a randomized complete block design. Table 1.2 shows relative efficiencies (to two decimal places) for different values of ρ in (1.3.16).

Table 1.1 Example of a First-Order NN Balanced Complete Block Design

Blocks				
1	2	3	4	5
1	2	0	3	9
2	3	1	4	0
0	1	9	2	8
3	4	2	5	1
9	0	8	1	7
4	5	3	6	2
8	9	7	0	6
5	6	4	7	3
7	8	6	9	5
6	7	5	8	4

Source: Grondona and Cressie (1992).

Table 1.2 Relative Efficiencies of the First-Order NN Balanced Design Given in Table 1.1 (A_O/A_N), and a Randomized Complete Block Design (A_O/A_R)

	ρ						
	−0.75	−0.50	−0.25	0	0.25	0.50	0.75
A_O/A_N	1.00	1.00	1.00	1.00	1.00	1.00	1.00
A_O/A_R	0.33	0.67	0.92	1.00	0.87	0.63	0.36

Source: Grondona and Cressie (1992).

Optimal designs may not exist; first-order NN balanced designs are relatively easier to construct and appear to be (almost) as efficient as the optimal designs. Further, from Table 1.2, randomized complete block designs can be grossly inefficient when $|\rho| \geq 0.5$. More general spatial correlation structures are considered in Grondona and Cressie (1992) and Section 5.6.2; the same efficiency patterns emerge.

There is often a big difference between what scientists and engineers *should* do to obtain statistical optimality and what they do in practice. Randomized complete block designs may be inefficient, but they are very, very easy to construct. In Section 5.7, it is shown how to recover some of the lost efficiency after the inefficient (nonspatial) design is used. My preference is for a spatial modeling approach, such as the one described in Section 5.7.2 and in Grondona and Cressie (1991): The strategy there is to estimate and model the spatial correlation from (median-polish) residuals, and then to use generalized-least-squares estimates of the treatment effects $\{\tau_k: k = 1, \ldots, t\}$. An artificial experiment is constructed by adding treatment effects (according to a randomized complete block design) to spatially correlated uniformity data; the spatial analysis yields gains in efficiency of over 30%.

In many applications of Statistics to problems that are inherently spatial, data are observational rather than the result of a controlled experiment. Thus, the opportunity to randomize, block, and replicate is often not available. In these circumstances, the model fitted to the data may implicitly assume that some uncontrolled-for effects or interactions are zero. Hopefully, such assumptions are based on prior knowledge of the phenomenon and not on mathematical convenience. Wiens et al. (1986) report an interesting seven-year study of territorial locations, territorial sizes, and breeding densities within bird-breeding areas in central Oregon. They manipulated the habitat compostion and structure by removing 75%, 50%, 25%, and 0% of the shrubs from 625-m^2 blocks in a checkerboard design. A nearby, unaltered area served as the control. In spite of the elaborate experiment, the authors expressed concern about uncontrollable factors such as site tenacity by breeding adults and about the appropriateness of the spatial scale on which the manipulation was conducted. In fact, these same concerns are relevant to *all* designed experiments; it is simply a question of degree.

Linear Models with Spatially Dependent Error

Suppose the spatial data $Z(\mathbf{s}_1), \ldots, Z(\mathbf{s}_n)$, observed at spatial locations $\{\mathbf{s}_1, \ldots, \mathbf{s}_n\}$, are modeled as a collection of random variables, generated by the random process

$$Z(\mathbf{s}) = \sum_{l=1}^{q} \beta_l x_l(\mathbf{s}) + \delta(\mathbf{s}), \qquad \mathbf{s} \in D \subset \mathbb{R}^d, \tag{1.3.18}$$

where $\{x_l(\cdot)\colon l = 1, \ldots, q\}$ is a collection of q nonrandom explanatory variables that may or may not depend on spatial location and $\delta(\cdot)$ is a zero-mean finite-variance error process that may or may not be spatially correlated.

For example, the model with constant mean μ and covariances given by (1.3.3) is a special case of (1.3.18) in $\mathbb{R}^1$, with just one explanatory variable $x_1(\cdot) \equiv 1$, $\beta_1 \equiv \mu$, and $\operatorname{cov}(\delta(u), \delta(v)) = \sigma_0^2 \cdot \rho^{|u-v|}$, $u, v \in \mathbb{R}^1$. Also, the complete-block-design model (1.3.15) and (1.3.16) is a special case of (1.3.18) in $\mathbb{R}^2$, where explanatory variables are $x_1(\cdot) \equiv 1$, $x_{k+1}(i, j) = 1$ if the (i, j)th plot receives the kth treatment and zero otherwise $(k = 1, \ldots, t)$, and $x_{l+t+1}(i, j) = 1$ if $j = l$ and zero otherwise $(l = 1, \ldots, b)$. The error process has anisotropic spatial correlation given by (1.3.16), specifying correlation within blocks but not between blocks.

It is helpful to write the observations as $\mathbf{Z} \equiv (Z(\mathbf{s}_1), \ldots, Z(\mathbf{s}_n))'$ and the model given by (1.3.18) in matrix and vector notation. That is,

$$\mathbf{Z} = X\boldsymbol{\beta} + \boldsymbol{\delta}, \tag{1.3.19}$$

where X is an $n \times q$ matrix whose (i, j)th element is $x_j(\mathbf{s}_i)$, $\boldsymbol{\beta} \equiv (\beta_1, \ldots, \beta_q)'$, and $\boldsymbol{\delta} \equiv (\delta(\mathbf{s}_1), \ldots, \delta(\mathbf{s}_n))'$.

Often (although not always), the parameters of greatest interest are the "large-scale-variation" parameters $\boldsymbol{\beta}$. For the moment, assume that any "small-scale-variation" parameters associated with the error process $\delta(\cdot)$ are known. That is, one can obtain

$$\operatorname{var}(\boldsymbol{\delta}) = \Sigma, \tag{1.3.20}$$

where Σ is a known $n \times n$, symmetric, nonnegative-definite matrix.

To keep the exposition simple, in all that is to follow, I shall assume that the regression coefficients $\boldsymbol{\beta}$ are estimable, Σ is invertible, and matrix inverses exist whenever I need them. Generalizations of this simple paradigm can be found, for example, in Searle (1971), Section 5.8.

Suppose the goal is efficient *estimation* of $\boldsymbol{\beta}$. [Efficient *prediction* of an unknown $Z(\mathbf{s}_0)$ is another possible goal; see Chapter 3.] The best linear unbiased estimator of $\boldsymbol{\beta}$ is also the *generalized-least-squares* (g.l.s.) estimator

(Aitken, 1935), that is, the value of β that minimizes

$$(\mathbf{Z} - X\beta)'\Sigma^{-1}(\mathbf{Z} - X\beta). \tag{1.3.21}$$

It is

$$\hat{\beta}_{gls} = (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}\mathbf{Z}, \tag{1.3.22}$$

which is also the maximum likelihood estimator if $\delta(\cdot)$ is a Gaussian process. The variance matrix of (1.3.22) is

$$\text{var}(\hat{\beta}_{gls}) = (X'\Sigma^{-1}X)^{-1}. \tag{1.3.23}$$

When $\Sigma = \sigma^2 I$, which it is under classical (nonspatial) assumptions of i.i.d. errors, (1.3.21) reduces to $(\mathbf{Z} - X\beta)'(\mathbf{Z} - X\beta)/\sigma^2$. Then the appropriate minimizer is

$$\hat{\beta}_{ols} = (X'X)^{-1}X'\mathbf{Z}, \tag{1.3.24}$$

which is the *ordinary-least-squares* (o.l.s.) estimator of β. However, when $\Sigma \neq \sigma^2 I$ [e.g., Σ has entries given by (1.3.16)], the o.l.s. estimator is often inefficient. Specifically,

$$\text{var}(\hat{\beta}_{ols}) = (X'X)^{-1}(X'\Sigma X)(X'X)^{-1} \tag{1.3.25}$$

and the difference $\text{var}(\hat{\beta}_{ols}) - \text{var}(\hat{\beta}_{gls})$ is nonnegative-definite (e.g., Searle, 1971, Section 3.3). Notice that when $\Sigma = \sigma^2 I$, the difference is 0.

Because the o.l.s. estimator (1.3.24) does not require knowledge of Σ, considerable research effort has been devoted to finding conditions on X and Σ for which o.l.s. and g.l.s. are equally efficient or for which o.l.s. and g.l.s. estimators are the same. Zyskind (1967) derives eight equivalent necessary and sufficient conditions for the latter to happen; for example, one such condition is $X(X'X)^{-1}X'\Sigma = \Sigma X(X'X)^{-1}X'$, which occurs if and only if Σ is of the form $a^2 I + P_1 A P_1 + P_2 B P_2$, where $P_1 = X(X'X)^{-1}X'$ and $P_2 = I - P_1$.

However, care should be taken. Just because o.l.s. and g.l.s. yield the same estimators, it cannot be assumed that all inference can proceed as if $\Sigma = \sigma^2 I$. Indeed, under Zyskind's conditions, $\text{var}(\hat{\beta}_{ols}) = \text{var}(\hat{\beta}_{gls}) = (X'\Sigma^{-1}X)^{-1}$, which depends on Σ. Therefore, from a total-inference point of view, results on the equivalence of o.l.s. and g.l.s. are not all that helpful. Moreover, except for a few special cases, spatial linear models like (1.3.18) generally give rise to matrices X and Σ that do *not* satisfy Zyskind's conditions. [Kramer and Donninger, 1987, give one of these special cases when $\delta(\cdot)$ is a spatial autoregressive process of the type described in Section 6.7.2. Curiously, they show that o.l.s. can be as *efficient* as g.l.s. when the spatial-

dependence parameter in $\delta(\cdot)$ tends to its largest possible value.] The generally poor performance of classical statistical-inference procedures in the presence of spatial dependence (see, e.g., the simple examples above; Cliff and Ord, 1981, Chapter 7; Dow et al., 1982; Anselin and Griffith, 1988) and the ubiquity of such spatial dependence, demonstrate why Statistics for spatial data should be part of every statistician's repertoire.

Valid Spatial Models

In reality, Σ is unknown; typically it will depend on parameters $\boldsymbol{\theta}$ obtained from a spatial parametric covariance model $C(\mathbf{u}, \mathbf{v}; \boldsymbol{\theta}) \equiv \text{cov}(\delta(\mathbf{u}), \delta(\mathbf{v}))$, $\mathbf{u}, \mathbf{v} \in D \subset \mathbb{R}^d$. For example, assuming homoskedasticity, one parameter will certainly be $\sigma^2 \equiv \text{var}(\delta(\mathbf{u}))$. Let $\Sigma(\boldsymbol{\theta})$ denote the $n \times n$ variance matrix var($\mathbf{Z}$) whose (i, j)th element is $C(\mathbf{s}_i, \mathbf{s}_j; \boldsymbol{\theta})$.

Choice of an appropriate model can be a difficult task because one must take care to use valid models; that is, use only those $\boldsymbol{\theta}$ for which $\sum_{i=1}^{m}\sum_{j=1}^{m} a_i C(\mathbf{u}_i, \mathbf{u}_j; \boldsymbol{\theta}) a_j \geq 0$, for *any* m, *any* sequence of real numbers $\{a_i: i = 1, \ldots, m\}$, and *any* sequence of spatial locations $\{\mathbf{u}_i: i = 1, \ldots, m\}$. For example, a valid covariance model in $\mathbb{R}^1$ is $C(s_i, s_j; \theta) = C_T(|s_i - s_j|; \theta)$, the "tent" covariogram

$$C_T(h; \theta) \equiv \begin{cases} \sigma^2\left(1 - \dfrac{|h|}{\theta}\right), & 0 \leq |h| \leq \theta, \\ 0, & |h| > \theta. \end{cases}$$

However, in $\mathbb{R}^2$, $C(\mathbf{s}_i, \mathbf{s}_j; \theta) \equiv C_T(\|\mathbf{s}_i - \mathbf{s}_j\|; \theta)$ is *not* a valid model; see Section 2.5.1 for a counterexample. On the other hand, if the spatial model for data $\mathbf{Z} = (Z(\mathbf{s}_1), \ldots, Z(\mathbf{s}_n))'$ has sufficient measurement error so that $\Sigma(\boldsymbol{\theta})$ is diagonally dominant, that is, if

$$C(\mathbf{s}_i, \mathbf{s}_i; \boldsymbol{\theta}) \geq \sum_{\substack{j=1 \\ j \neq i}}^{n} \left|C(\mathbf{s}_i, \mathbf{s}_j; \boldsymbol{\theta})\right|, \qquad i = 1, \ldots, n,$$

then $\Sigma(\boldsymbol{\theta})$ is always nonnegative-definite (Varga, 1962, p. 22ff) and hence a valid model is obtained. In the preceding simple examples, the models (1.3.3) and (1.3.16) are valid provided the parameter $\rho \in (-1, 1)$.

Questions of validity of the various spatial models presented in this book will be dealt with as they arise. It is important to ensure that the parameter space Θ of $\boldsymbol{\theta}$ consists only of valid models, and that an estimate $\hat{\boldsymbol{\theta}}$ of $\boldsymbol{\theta}$ belongs to Θ.

Estimated Generalized Least Squares

Assume that the spatial dependence parameter $\boldsymbol{\theta}$ is contained in a known parameter space Θ, any member of which yields a valid spatial model. Any estimator $\hat{\boldsymbol{\theta}}$ will be assumed to belong to Θ, with probability 1.

Estimated generalized-least-squares (e.g.l.s.) estimators are obtained by substituting $\Sigma(\hat{\boldsymbol{\theta}})$ for $\Sigma(\boldsymbol{\theta})$ in (1.3.22):

$$\hat{\boldsymbol{\beta}}_{\text{egls}} = \left(X'\Sigma(\hat{\boldsymbol{\theta}})^{-1}X\right)^{-1}X'\Sigma(\hat{\boldsymbol{\theta}})^{-1}\mathbf{Z}, \tag{1.3.26}$$

where $\hat{\boldsymbol{\theta}}$ is an estimator of the small-scale-variation parameters $\boldsymbol{\theta}$.

Prediction of an unknown $Z(\mathbf{s}_0)$ proceeds in a similar way, often by assuming that the estimated value $\hat{\boldsymbol{\theta}}$ is the true value (Chapter 3). The consequences of this assumption are investigated in Section 5.3.

Two different approaches are often employed to obtain $\hat{\boldsymbol{\theta}}$. If the probability density (or mass) function of $\mathbf{Z}$ is completely specified as $f(\mathbf{z}; \boldsymbol{\beta}, \boldsymbol{\theta})$, then the likelihood

$$l(\boldsymbol{\beta}, \boldsymbol{\theta}) \equiv f(\mathbf{Z}; \boldsymbol{\beta}, \boldsymbol{\theta})$$

can be maximized with respect to $\boldsymbol{\beta}$ and $\boldsymbol{\theta}$ to yield estimators $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\theta}}$. Then (1.3.26) is the maximum likelihood (m.l.) estimator of $\hat{\boldsymbol{\beta}}_{\text{gls}}$. Sections 7.2 and 7.3 deal mainly with this approach.

A second possibility is to obtain an initial estimate of $\boldsymbol{\beta}$ (e.g., $\hat{\boldsymbol{\beta}}_{\text{ols}}$), construct residuals

$$\mathbf{R} \equiv \mathbf{Z} - X\hat{\boldsymbol{\beta}}_{\text{ols}}, \tag{1.3.27}$$

and, using these residuals as a proxy for $\boldsymbol{\delta}$, estimate $\boldsymbol{\theta}$ free from the presence of $\boldsymbol{\beta}$. This method is often employed in a semiparametric setting where the joint distribution of $\mathbf{Z}$ is not specified, but the mean $X\boldsymbol{\beta}$ and the variance $\Sigma(\boldsymbol{\theta})$ are specified. Then, $\hat{\boldsymbol{\theta}}$ can be obtained from an approximate method-of-moments approach based on $\mathbf{R}$ (Section 2.6.2). From $\hat{\boldsymbol{\theta}}$ and (1.3.26), $\hat{\boldsymbol{\beta}}_{\text{egls}}$ is obtained. One then has the option of substituting this estimate back into (1.3.27) to obtain a new $\mathbf{R}$, a new $\hat{\boldsymbol{\theta}}$, and a new $\hat{\boldsymbol{\beta}}_{\text{egls}}$, and so forth. Terminating at this second step yields a two-stage estimation procedure.

The example considered throughout this section [$\mathbf{Z}$ Gaussian with covariances defined by (1.3.3)] satisfies the conditions of Mardia and Marshall's (1984) theorem given in Section 7.3.1. Hence, for $\boldsymbol{\theta} = (\sigma^2, \rho)'$ and $\hat{\boldsymbol{\theta}}$ its m.l. estimator, $\hat{\boldsymbol{\beta}}_{\text{egls}}$ given by (1.3.26) is asymptotically Gaussian with mean $\boldsymbol{\beta}$ and variance matrix $(X'\Sigma(\boldsymbol{\theta})^{-1}X)^{-1}$ and is asymptotically independent from $\hat{\boldsymbol{\theta}}$. However, for small sample sizes, the m.l. estimators of the small-scale-variation parameters are notoriously biased. A review of results for the model (1.3.3) can be found in Dielman and Pfaffenberger (1989), together with small-sample simulations and recommendations.

The exact variance of $\hat{\boldsymbol{\beta}}_{\text{egls}}$ is difficult to compute because $\mathbf{Z}$ appears in (1.3.26) nonlinearly. Preliminary tests on $\boldsymbol{\theta}$ (see Section 7.3.1) are often employed to determine whether g.l.s. (actually, e.g.l.s.) will be used in preference to o.l.s. Intuitively, as sample size increases, e.g.l.s. will be indistinguishable from g.l.s., provided $\hat{\boldsymbol{\theta}}$ is a consistent estimator of $\boldsymbol{\theta}$.

Williams (1975) can be consulted for results that formalize this intuition. In finite samples, terms of smaller orders of magnitude can improve the asymptotic distribution of $\hat{\boldsymbol{\beta}}_{egls}$ (e.g., Rothenberg, 1984; Grubb and Magee, 1988). Asymptotics in spatial problems can be of the infill type (Sections 2.6 and 5.8) or of the increasing-domain type (Section 7.3). Statisticians and probabilists are most familiar with the latter; indeed, all of the asymptotic results mentioned in this section are for increasing domains.

Some interesting finite-sample results that establish inequalities between $E(\widehat{\text{var}}(\mathbf{c}'\hat{\boldsymbol{\beta}}_{gls}))$, $\text{var}(\mathbf{c}'\hat{\boldsymbol{\beta}}_{gls})$, and $\text{var}(\mathbf{c}'\hat{\boldsymbol{\beta}}_{egls})$ are proved by Harville (1985) and Eaton (1985). For example, Harville shows that if $\hat{\boldsymbol{\theta}}$ is an even, translation-invariant estimator, and the data are Gaussian, then

$$\text{var}\left(\mathbf{c}'\hat{\boldsymbol{\beta}}_{gls}\right) \leq \text{var}\left(\mathbf{c}'\hat{\boldsymbol{\beta}}_{egls}\right), \tag{1.3.28}$$

for all $\mathbf{c} \in \mathbb{R}^q$. Using a matrix form of Jensen's inequality, Eaton shows that if $E(\Sigma(\hat{\boldsymbol{\theta}})) = \Sigma(\boldsymbol{\theta})$, then

$$E\left(\widehat{\text{var}}\left(\mathbf{c}'\hat{\boldsymbol{\beta}}_{gls}\right)\right) \leq \text{var}\left(\mathbf{c}'\hat{\boldsymbol{\beta}}_{gls}\right), \tag{1.3.29}$$

where $\widehat{\text{var}}(\mathbf{c}'\hat{\boldsymbol{\beta}}_{gls})$ is $\mathbf{c}'(X'\Sigma(\hat{\boldsymbol{\theta}})^{-1}X)^{-1}\mathbf{c}$, the often used estimator of $\text{var}(\mathbf{c}'\hat{\boldsymbol{\beta}}_{gls})$; see (1.3.23). Although for $\hat{\boldsymbol{\theta}}$ a consistent estimator of $\boldsymbol{\theta}$ the three quantities in (1.3.28) and (1.3.29) are asymptotically indistinguishable, the inequalities show that, as an estimator of $\text{var}(\mathbf{c}'\hat{\boldsymbol{\beta}}_{egls})$, $\widehat{\text{var}}(\mathbf{c}'\hat{\boldsymbol{\beta}}_{gls})$ has potentially two sources of negative bias. Harville and Jeske (1992) investigate estimators of $\text{var}(\mathbf{c}'\hat{\boldsymbol{\beta}}_{egls})$ that attempt to correct for this bias.

Spatial Data Analysis and Modeling: A Summary

- All data have a more-or-less precise spatial and temporal label associated with them. A purely spatial model usually has no causative component in it; such models are useful when a space–time process has reached temporal equilibrium (e.g., ore deposition), or when short-term causal effects are aggregated over a fixed time period (e.g., final presidential election returns from the states of the United States).
- Whether the spatial labels are thought to be an important part of the modeling and analysis of the data is a concern that should be addressed problem by problem.
- Data that are close together in space (and time) are often more alike than those that are far apart. A spatial model incorporates this spatial variation into the generating mechanism, in contrast to a nonspatial model.
- It is almost always true that the classical, nonspatial model is a special case of a spatial model, and so the spatial model is more general (space–time models are even more general).

- Whether one chooses to model the spatial variation through the (nonstochastic) mean structure (called large-scale variation throughout the book) or the stochastic-dependence structure (called small-scale variation) depends on the underlying scientific problem, and can sometimes be simply a trade-off between model fit and parsimony of model description. What is one person's (spatial) covariance structure may be another person's mean structure.
- Explanatory variables (suggested by the problem under investigation) should be included in the mean structure first, and great care should be taken to find all of them. A missed variable that is itself varying spatially will contribute to the spatial dependence, as can a misspecification of the functional relationship between the independent and explanatory variables. As a consequence, a model that includes a spatial-dependence component pays a low-cost premium that insures against misspecification of the mean structure (e.g., Dubin, 1988).
- Having allowed for explanatory variables, models with spatial dependence typically have a more parsimonious description than classical trend-surface models. They also have more stable spatial extrapolation properties and yield more efficient estimators of explanatory-variable effects.

Some Final Remarks

The simple examples presented in this chapter serve to illustrate a general principle in all of Statistics, namely that inferences can depend crucially on assumptions. In situations such as for spatial or temporal data, where it is often more realistic to abandon the independence assumption, classical inferences can be very misleading. Stephan (1934) saw this very clearly in problems dealing with social data; he wrote

> ... Data of geographic units are tied together, like bunches of grapes, not separate, like balls in an urn. Of course, mere contiguity in time and space does not of itself indicate lack of independence between units in a relevant variable or attribute, but in dealing with social data, we know that by virtue of their very *social* character, persons, groups and their characteristics are interrelated and not independent. Sampling error formulas may yet be developed which are applicable to these data, but until then the older formulas must be used with great caution. Likewise, other statistical measures must be carefully scrutinized when applied to these data ... .

In the last 40 years, considerable attention has been devoted to concerns of the type expressed by Stephan; see Section 5.6.1 for a brief review.

Among the exact sciences, it is often believed that the only source of variability is measurement error, which is usually modeled as a white-noise random process. In my opinion, this viewpoint is far too simplistic, and could lead to important small-scale (spatial or nonspatial) effects being missed.

Problems in the earth sciences, ecology, soil science, hydrology, and so on can provide the statistician with a considerable challenge. Often studies are observational rather than designed, and there is no replication because there is just one unit: the earth! Nevertheless, explanations for various phenomena can be conjectured and their consistency across time or space can be assessed. But, what is the source of random variation in the spatial models (and temporal models, for that matter) proposed? A standard tactic in science is that when there are infinitely many possible data values from which to choose, those possibilities are dealt with statistically. With no data, but with experience in the way, say, oilfields of a particular type behave, one might assume that a spatial model with trend and additive stationary error would be appropriate.

Suppose that you are confronted with a spatial data set and a scientific or engineering problem to solve. After some *descriptive* analyses of the data, the set of potential models can hopefully be narrowed to just a few. An *indication* of a model's parameters can be obtained by applying an estimation criterion, such as maximum likelihood, maximum pseudolikelihood, method-of-moments, and so forth. Whether these fitted parameters are significant is determined by statistical-inferential methods. *Estimation* of a model's parameters involves both indication and, at least, measures of the estimator's variation. In either case (indication or estimation), it is important to check the fit of the model (because no model is exactly true) using *diagnostic* statistics and plots.

The approach taken in this book is to broaden the classical models and methods to those that recognize the presence and importance of spatial information. The success of these spatial models and methods will be judged by their powers of organization, explanation, and prediction. Much has been accomplished in the last 20 years, and a vigorous future (in modeling, simulation, estimation, computation, inference, asymptotics, diagnostics, etc.) for the subject is assured.

A rereading of Section 1.1 would be helpful at this point, but for those who are ready to jump in, a brief synopsis is given as follows. Chapters 2, 3, and 4 deal with geostatistical data, and Chapter 5 is a collection of special topics of interest to me (Part I). Chapters 6 and 7 deal with lattice data (Part II). Chapters 8 and 9 deal with, respectively, point patterns and objects based on point patterns (Part III).

We shall not cease from exploration
And the end of all our exploring
Will be to arrive where we started
And know the place for the first time. (T. S. Eliot)